MILITARY SPECIFICATION

CDU-4(T-1)/B, CDU-5(T-1)/B, CDU-10(T-1)/B and CDU-4(T-2)/B

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the requirements for the loading (Practice APERS Mines, and Inert Anti-Intrusion Devices) into two types of canisters and the assembling and packing of canisters.

1.1.1 Type I - Canister with explosive opener. These canisters are designated as

CDU-4(T-1)/B - Practice APERS Mines
CDU-5(T-1)/B - Inert Anti-Intrusion Device (ISBB's)
CDU-10(T-1)/B - Inert Anti-Intrusion Device (ISBB's & IEAID's)

1.1.2 Type II - Canister with training opener. This canister is designated as

CDU-4(T-2)/B - Practice APERS Mines

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

FSC: 1345

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MIL-R-11470	-	Radiographic Inspection, Qualification of Equipment, Operators and Procedures

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9227366	-	Packing and Marking for Kit, Mine Canister: CDU
9228096	-	CDU-4(T-1)/B and CDU-4(T-2)/B; Loading and Assembly
9228098	-	CDU-5/B, CDU-5(T-1)/B; Loading and Assembly
9228100	-	CDU-10/B and CDU-10(T-1)/B; Loading and Assembly
9229491	-	Subpack Loading Assembly with Drawing Mine XM62
9234095	-	CDU-4(T-1)/B and CDU-4(T-2)/B; Loading Assembly Procedure
9235168	-	CDU-5/B and CDU-5(T-1)/B Loading Assembly Procedure
9235641	-	CDU-10/B and CDU-10(T-1)/B Loading Assembly Procedure

PUBLICATIONS

ARMY

ET 9228096	-	CDU-4 (T-1)/B & CDU-4(T-2)/B Loading & Assembly
ET 9228098-2	-	CDU-5(T-1)/B Loading & Assembly
ET 9228100-2	-	CDU-10(T-1)/B Loading & Assembly

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	-	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 71-78 (latest revisions)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components

3.2.1 General.-The components shall comply with all requirements specified on applicable drawings and specifications.

3.2.2 Continuity of Liquid Level Sensor, (dwg. 9227930) prior to assembly.-The liquid level sensor shall have electrical continuity of no more than 0.15 ohms resistance from pin E to pin H of the connector when the float is in its uppermost vertical position. The continuity and resistance (pin E to pin H) shall be continuous during the downward travel of the float for 0.062" minimum. After the float has dropped at least 0.062" from its uppermost position and before it reaches its bottommost position, the sensor shall show continuity of no more than 0.15 ohms resistance from pin E to pin D of the connector.

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The continuity and resistance (pin E to pin D) shall be continuous for the remaining float drop. As the float rises and before it reaches 0.062" from its uppermost position, the sensor shall again show continuity of no more than 0.15 ohms resistance from pin E to pin H. The continuity and resistance (pin E to pin H) shall be continuous as the float rises to its uppermost position. At no time shall there be simultaneous continuity of both (pin E to pin H) and (pin E to pin D). The resistance of each open circuit shall be tested and must be greater than 1 megohm.

3.2.3 Shunt verification of explosive opener fixed delay assembly (dwg. 9233359) (Type I only).-There shall be electrical continuity and no more than 0.3 ohms resistance measured at the termination of the detonator leads.

3.2.4 Continuity, resistance and insulation resistance of the cable assembly (dwg. 9223682) (prior to assembly).-There shall be continuity and no more than 0.1 ohm resistance between each pair of wired contacts of the cable assembly. There shall also be a minimum of 10 megohms resistance between each wire and all other wires of the P1 connector.

3.2.5 Functioning of the electronic opener assembly (dwg. 9227463) (prior to assembly)

3.2.5.1 Minimum time delay.-The opener shall show a time delay of 1.25 seconds $\pm$ 0.25 seconds before exhibiting an output pulse. When its capacitors are charged as specified in 4.4.4.

3.2.5.2 Maximum time delay.-The opener shall show a time delay of 24.00 seconds $\pm$ 2.40 seconds before exhibiting an output pulse. When its capacitors are charged as specified in 4.4.4.

3.2.5.3 Output.-The opener shall exhibit a minimum output of one (1) volt for one hundred (100) microseconds measured across a two (2) ohm + 0.20 ohm load.

3.2.6 Continuity and resistance of the opener switch assembly (dwg. 9227439) prior to assembly

3.2.6.1 With switches in normally closed (relaxed) position.-The opener switch assembly shall show evidence of continuity and no more than 10.0 ohms resistance between the B pin of the

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connector and the terminal closest to the chamfered edge of the plug (blue wires) and between pin D of the connector and the terminal farthest from the chamfered edge of the plug (yellow wires). There shall also be no more than 0.5 ohm resistance between pin C of the connector and the pin receptacle (black wires).

3.2.6.2 With switches actuated.-There shall be less than 600 K ohms between pin C of the connector and the terminal closest to the chamfered edge of the plug and less than 60 K ohms between pin C of the connector assembly and the terminal farthest from the chamfered edge of the plug. A maximum force of 1000 grams shall be applied to the switch plungers to cause them to actuate.

3.3 Assembly.-The CDU-Assembly shall comply with all requirements specified on applicable drawings (dwgs) and with all requirements specified on applicable specifications.

3.4 Desensitizer (TCTFE) leakage.-The loaded and sealed canister shall exhibit no desensitizer (TCTFE) leakage in excess of 1.0×10^{-3} cc/sec.

3.5 Continuity of the electrical circuits of the loaded canister assembly.-There shall be continuity between the following points:

a. Pin 4 and Pin 5 of the P1 connector.
b. Pin 7 of the P1 connector and the electronic opener mounting plate.
c. There shall be less than 600 K ohms resistance between the opener mounting plate and the terminal closest to the chamfered edge of the opener switch assembly plug.
d. There shall be less than 60 K ohms resistance between the opener mounting plate and the terminal farthest from the chamfered edge of the opener switch assembly plug.

There shall be no continuity between the following points:

e. Pin 6 of the P1 connector and the terminal closest to the chamfered edge of the opener switch assembly plug.
f. Pin 8 of the P1 connector and the terminal farthest from the chamfered edge of the opener switch assembly plug.

For purposes of this requirement, no continuity shall mean a resistance greater than 10 megohms.

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3.6 Assembly (CDU Mine Canister Kit).-The CDU Mine Canister Kit shall comply with all requirements specified on dwg. 9227366 and with all requirements specified in applicable specifications.

3.7 Continuity of loaded container.-The loaded container shall show evidence of electrical continuity when tested through the monitor and ready connector (P2 connector).

3.8 Tension verification of strapping machine.-Strap tension shall be 140 pounds, min., when tested as specified in 4.4.9. (Stall reading, prior to engaging weld level, shall not exceed 250 pounds.)

3.9 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.10 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-105 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. For type I, the first article sample shall consist of twenty (20) canisters (two kits) of each canister to be produced in accordance with the contract. For Type II, the first article sample shall consist of ten (10) canisters (one kit) of each canister to be produced in accordance with the contract. All parts and completed assemblies submitted, shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article sample or portion thereof, as

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directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period of 90 days, or whenever a change occurs in source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies will be subjected to the following tests:

4.2.2.1 Desensitizer (TCTFE) leakage (see 3.4).-The sample canisters shall be subjected to this test. The test shall be performed as specified in 4.4.6. This test is considered non-destructive.

4.2.2.2 Continuity of the electrical circuits of the loaded canister assembly (see 3.5).-The sample canisters shall be subjected to this test. The test shall be performed as specified in 4.4.7. This test is considered non-destructive.

4.2.2.3 Continuity of loaded container (see 3.7).-The sample containers (kits) shall be subjected to this test. The test shall be performed as specified in 4.4.8. This test is considered non-destructive.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample

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is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Mines of one lot interfix number from one manufacturer.
b. Canister of one lot interfix number from one manufacturer.
c. Explosive opener fixed delay assemblies of one lot interfix number from one manufacturer. (Type I only.)
d. Electronic openers and switch assemblies of one lot interfix number from one manufacturer.
e. Liquid level sensors of one lot interfix number from one manufacturer.
f. Fillers and plugs of one lot interfix number from one manufacturer.
g. Caps of one lot interfix number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all

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of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Canister, prior to loading (see applicable dwg.)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		
Major	AQL 100 Percent		
101.	Protective sleeve missing or improperly assembled (Type I only)	Visual/Manual	01001
102.	Subpack flap secured (CDU-10(T-1)/B only)	Visual	01002
103.	Hinge pin does not rotate 360° (CDU-10(T-1)/B only)	Manual	01003
Major	AQL 1.00 Percent		
104.	EOD tube damaged, missing or improperly assembled	Visual	01004
105.	Location of hole for indicator eye	Gage	01005
106.	Filler improperly assembled	Visual/Manual	01006
107.	Depth of counterbore for indicator eye	Gage	01007
108.	Flushness of the tube assembly	Gage	01008
109.	Protrusion of indicator eye, max.	Gage	01009
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.9)	Visual	01010

4.3.2.2 Canister, after sealing and prior to final assembly (see applicable dwg.)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		

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Major:	AQL 100 Percent		
101.	Incorrect torque of liquid level sensor	Test	02001
102.	Loading improperly performed	Visual	02002
Major:	AQL 1.00 Percent		
103.	Alignment of cap assembly	Gage	02003
104.	Depth of cap assembly	Gage	02004
105.	Parallelism of cap assembly	Gage	02005
106.	Excessive heat seal flashing	Visual	02006
107.	Plug not properly assembled	Visual	02007
108.	Torque of base plate mounting screws	Test	02008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.9)	Visual	02009

4.3.2.3 Canister Assembly (See Applicable Dwg.)

Categories	Defects	Method of Inspection	Defect Code (see 6.2)
Critical:	None defined		
Major:	AQL 100 Percent		
101.	Distance to EOD marking from top of canister, max.	Gage	03001
102.	Color of EOD marking (2 places)	Visual	03002
103.	Lanyard damaged, broken or improperly assembled	Visual	03003
104.	Plastic connectors between the opener and switch assembly not mated (prior to attaching the lid assembly and after performing the test in 4.4.7)	Visual	03004
105.	Pin receptacle not securely fastened to opener mounting plate ground pin (prior to attaching lid assembly and after performing the test in 4.4.7)	Visual/Manual	03005
106.	Incorrect number of opener plate spacers	Visual	03006

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107.	Gap between opener mounting plate and support adapter	Gage	03007
108.	Gap between opener switch mounting plate and opener mounting plate	Gage	03008
109.	Improper routing of electrical special purpose cable assembly and electronic opener cable	Visual	03009
110.	Staking of retention rods and stamped nuts missing or improper	Visual	03010
111.	Explosive opener cable assembly pinched between base plate and opener plate or between opener plate and electronic opener	Visual/Manual	03011
112.	Loose, cut or exposed wire or lanyards (after switch assembly, before and after cover assembly)	Visual	03012
113.	Flushness of explosive opener	Gage	03013
114.	Explosive opener not tightly seated	Manual	03014
115.	Lead wires tied to lanyard (just prior to assembly of cover assembly)	Manual	03015
Major:	AQL 1.00 Percent		
116.	Torque of retention rod nuts	Test	03016
117.	Torque of electronic opener and opener switch assembly mounting screws	Test	03017
118.	Electronic opener cable and explosive opener cable improperly mated and clamped	Visual	03018
119.	Any component missing or damaged	Visual	03019
120.	Marking incorrect, misleading or unidentifiable	Visual	03020
121.	Feed-thru splice missing or loose	Visual/Manual	03021
122.	Cargo door assembled incorrectly (door plug not aligned with EOD tube)	Visual	03022
123.	Torque of lanyard retaining screw	Test	03023

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124.	Improper positioning of straps and windshield pads	Visual/Manual	03024
125.	Alignment of cargo door, retention rods and retention ring relative to canister	Gage	03025
126.	Torque of cargo door screws	Gage	03026
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	03027

4.3.2.4 Container for CDU, prior to assembling crate cover (see dwg. 9227366)

Categories	Defects	Method of Inspection	Defect Code (see 6.2)
Critical:	None defined		
Major:	AQL 100 Percent		
101.	Monitor cable connector(s) assembled to canister(s) in incorrect bay(s) (see 4.4.8)	Visual	04001
102.	Canisters improperly placed in crate	Visual	04002
103.	EOD entry holes unclear	Visual	04003
104.	Ejector assembly package missing or improperly secured to platform	Visual	04004
105.	Pressure cartridge assembly package missing or improperly secured to platform	Visual	04005
Minor:	AQL 0.65 Percent		
201.	Hardware and wiring of assembled canister(s) not matching bay number(s) of container that canister is placed in	Visual	04006
202.	Improper position of special purpose electrical cable assembly	Visual	04007
203.	Evidence of poor workmanship (See 3.9)	Visual	04008

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4.3.2.5 Container for CDU (see dwg. 9227366)

Categories	Defects	Method of Inspection	Defect Code (see 6.2)
Critical:	None defined		
Major:	AQL 100 Percent		
101.	Indicator eye cannot be read or indicates a low desensitizer level	Visual	05001
Major:	AQL 0.65 Percent		
102.	Torque of screw(s) securing cover	Test	05002
103.	Marking incorrect, missing, misleading or unidentifiable	Visual	05003
104.	Safety panel(s) missing or damaged	Visual	05004
Minor:	AQL 1.00 Percent		
201.	Metallic seal(s) missing	Visual	05005
202.	Evidence of poor workmanship (See 3.9)	Visual	05006

4.3.3 Testing

4.3.3.1 Electrical continuity of liquid level sensor (see 3.2.2), Major Defect, Defect Code 06001.-Each liquid level sensor shall be subjected to this test prior to loading. If any sensor fails to comply with the applicable requirement, it shall not be used in assembly of the canister. The test shall be performed as specified in 4.4.1.

4.3.3.2 Shunt verification of explosive opener fixed delay assembly, prior to loading (see 3.2.3), Critical Defect, Defect Code 07001.-This test shall be performed 100 percent. Any explosive opener fixed delay assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.3 Electrical continuity, resistance and shorts of the cable assembly, prior to loading (see 3.2.4), Major Defect, Defect Code 09001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

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4.3.3.4 Functioning of electronic opener, prior to loading (see 3.2.5), Major Defect, Defect Code 08001.-This test shall be performed 100 percent. Any opener which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.5 Continuity and resistance of the opener switch assembly, prior to loading (see 3.2.6), Major Defect, Defect Code 08002.-A sample of 200 switches shall be selected from each lot and shall be tested as specified in 4.4.5, using equipment in accordance with 4.3.4. If any switch assembly fails to comply with the specified requirement, the lot shall be rejected.

4.3.3.6 Desensitizer (TCTFE) leakage (See 3.4), Major Defect, Defect Code 10001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4.

4.3.3.7 Continuity of the electrical circuits of the loaded canister (see 3.5), Major Defect, Defect Code 11001.-This test shall be performed 100 percent after assembly and re-torquing of the retention rod nuts. Any canister which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4.

4.3.3.8 Electrical continuity of loaded container (see 3.7), Major Defect, Defect Code 12001.-This test shall be performed 100 percent. Any loaded container which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

4.3.3.9 Tension verification of strapping machine, Major Defect, Defect Code 13001 (see 3.8).-Three (3) strap samples shall be welded and tested as specified in 4.4.9 to qualify each strapping machine for use in each production shift. In addition, one (1) sample per 160 welds shall be tested as a control of each machine during production. This test shall be performed using equipment in accordance with 4.3.4.

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If any strap sample fails during the qualification of the machine, that strapping machine shall not be used in production.

If the test sample used for control of the machine fails, three (3) additional samples shall be tested and if any sample fails, the machine shall be removed from production.

All strapping machines removed from production because of some faulty operation may, after rework, be returned to production provided they pass the above qualification test.

4.3.3.10 Leakage (see dwg. 9228100) Major Defect, Defect Code 14001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9225291 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Electrical continuity of liquid level sensor, prior to loading.-The liquid level sensor shall be tested using fixtures, instruments, and the method as shown on the equipment list and (dwg. 9242901). This is a non-destructive test; sensors that meet the requirements of 3.2.2 may be returned to the lot.

4.4.2 Shunt verification of explosive opener fixed delay assembly, prior to loading.-The equipment specified on the equipment lists shall be assembled to the explosive opener lead assembly and the opener assembly checked for compliance with the requirement specified in 3.2.3. A maximum current of one (1)

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milliampere shall be applied to the explosive opener. This is a non-destructive test; parts that meet the requirements of 3.2.3 may be returned to the lot.

WARNING: This test shall be performed behind a suitable barricade.

4.4.3 Electrical continuity, resistance and shorts of cable assembly, prior to loading.-The cable assembly shall be assembled to the equipment specified on the equipment list and checked for compliance with 3.2.4. This is a non-destructive test; parts that meet the requirements of 3.2.4 may be returned to the lot.

4.4.4 Functioning of electronic opener, prior to loading.- The electronic opener shall be assembled to a fixture as shown on the equipment list and tested for compliance with 3.2.5. This is a non-destructive test; parts that meet the requirements of 3.2.5 may be returned to the lot.

4.4.5 Continuity and resistance of the opener switch assembly, prior to loading.-The opener switch assembly shall be assembled to a fixture as shown on the equipment list and tested for compliance with 3.2.6. This is a non-destructive test; parts that meet the requirements of 3.2.6 may be returned to the lot.

4.4.6 Desensitizer (TCTFE) leakage (see 3.4).-The loaded canister shall be placed into a fixture as specified on the equipment list and tested for leakage. This is a non-destructive test; canisters that meet the requirements of 3.4 may be returned to the lot.

4.4.7 Continuity of the electrical circuit of the loaded canister (see 3.5).-The equipment specified on the equipment list shall be assembled to the loaded canister and the canister checked for compliance with the specified requirement. This test shall be performed after assembly and re-torquing of the retention rod nuts. This is a non-destructive test; canisters that meet the requirements of 3.5 may be returned to the lot.

4.4.8 Electrical continuity of the loaded container (see 3.7).- Prior to placing the cover on the container and connecting the monitor and test cable to the canisters, connect an XM67 Test Set (see dwg. 9221625) to the P2 receptacle mounted on the container. Insert the shorting plug specified on the equipment list into connector number 1 of the monitor and test cable. The monitor

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lite of the XM67 Test Set should light and the audible alarm should function. Remove the shorting plug and repeat the test until all ten (10) connectors have been checked. Remove the shorting plug (9227399) from the P1 receptacle mounted on the container and replace it with a ten-lite test set (see dwg. 9227398). Again insert the shorting plug mentioned above into connector number 1 of the monitor and test cable. Lite number 1 of the test set connector to the canister in the corresponding numbered bay. Repeat the test until all ten (10) connectors have been checked. Each connector shall be immediately mated to the canister in the numbered bay corresponding to the lighted lite on the ten-lite test set prior to checking the next connector. After all ten (10) canisters have been mated to the monitor and test cable, remove the ten-lite test set and replace it with the shorting plug (9227399). Depress the ready button on the XM67 Test Set and insure that the green ready lite lights. If the XM67 Test Set does not operate as stated above or if any lite on the ten-lite test set does not light when its corresponding numbered connector is shorted, the loaded container shall be rejected. This is a non-destructive test; containers that meet the requirements of 3.7 may be returned to the lot.

4.4.9 Strapping machine verification (see 3.8).-The test sample identified with the strapping machine being verified, shall be assembled to a fixture as specified on the equipment list and tested for strap tension. This test is considered destructive and samples upon completion of tests shall be destroyed.

4.4.10 Leakage test (see dwg. 9228100).-The canister with the indicator eye assembly and EOD tube assembly installed and prior to loading shall be placed into a fixture as specified on the equipment list and tested for leakage. This is a non-destructive test, canisters that meet the requirements of dwg. 9228100 may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Packing.-The loaded canister shall be packed in accordance with dwg. 9227366.

5.2 Marking.-Marking shall be in accordance with dwg. 9227366 and Interstate Commerce Commission Regulations 49 CFR 71-78.

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5.3 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of First Article Samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspection listed in this specification. These code numbers shall be listed on drawings of equipment designs and inspection data submitted by the contractor.

6.3 Inspection equipment.-The contractor shall design and submit for approval inspection equipment as required by the referenced Equipment Tabulation (ET) and each applicable Equipment List (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Tabulation.-This tabulation (SMUPA Form 1011) is an index of all applicable EL's.

6.3.2 Equipment Lists (EL).-This list (SMUPA Form 1010) indicates the availability of inspection equipment designs by showing in the "Part or Identifying No." column, the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.3. Design action required of the contractor is described in paragraphs 6.3.4 and 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.5 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

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6.3.3 Equipment list codes.-The inspection equipment as defined in 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor designs on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.4 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with an use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subjected to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three (3) copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.6.

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6.3.6 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND8. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of test results.-Information as required is Section 3 of this specification shall be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND8
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1345-A-269

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